

The statement $P \wedge Q$ is true. If $P \rightarrow Q$, then $\neg Q \rightarrow \neg P$. $P \leftrightarrow Q$ means $P \rightarrow Q \wedge Q \rightarrow P$.
 $\neg p \wedge q \wedge r \rightarrow s \rightarrow t$ The union $A \cup B$ of sets A and B . The complement $\overline{A}$ of set A . The power set $\mathcal{P}$
 (X) of set X . The cardinality $\#(A)$ of set A . $A \cup B \cap C \cap D \cap E \setminus F$ The union $\bigcup_{i=1}^n A_i$ of sets A_1
through A_n . The intersection $\bigcap_{j=0}^k B_j$ of sets B_0 through B_k . The set $\{x \mid x > 0\}$ of positive real
numbers. The set $\{n \mid n \in \mathcal{N} \wedge n \text{ is even}\}$ of even naturals.

Hi,

$p \wedge q$

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$\neg$

$\neg p$

1. Prove the following directly from the two definitions and three axioms given in ICA2 on 8/20. You may not use any other results without proving them first. Every statement in your proof must be fully justified. See **Elements of (Proof) Style** guidelines in Canvas for information.
 - (a) Direct proof: For all integers n , if n is odd, then $9n + 5$ is even.
 - (b) Contrapositive proof: For all integers n , if $7n + 5$ is odd, then n is even.
 - (c) Proof by contradiction: For all integers n , if $3n + 2$ is even, then n is even.